

WASP-43b

R-0403

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_250216 · created 2026-07-19T04:19:21+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460722.8977 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.169 +/- 0.015
Transit depth (Rp/Rs)^2	2.86 +/- 0.52 [%]
Orbital Inclination (inc)	83.08 +/- 1.55
Airmass coefficient 1 (a1)	1.001 +/- 0.015
Airmass coefficient 2 (a2)	-0.002 +/- 0.01
Scatter in the residuals of the lightcurve fit is	1.45 %
Best Comparison Star	#2 - [317, 69]
Optimal Method	PSF photometry
Transit Duration (day)	0.054 +/- 0.0042

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.169 ± 0.015 vs 0.1594 (deviation 0.64σ)
Duration fit vs published	0.054 d vs 0.0512 d (deviation 0.65σ)
Mid-time O–C	-4.2 min
Depth SNR	11.3
Residual scatter	1.45 %

Light curve

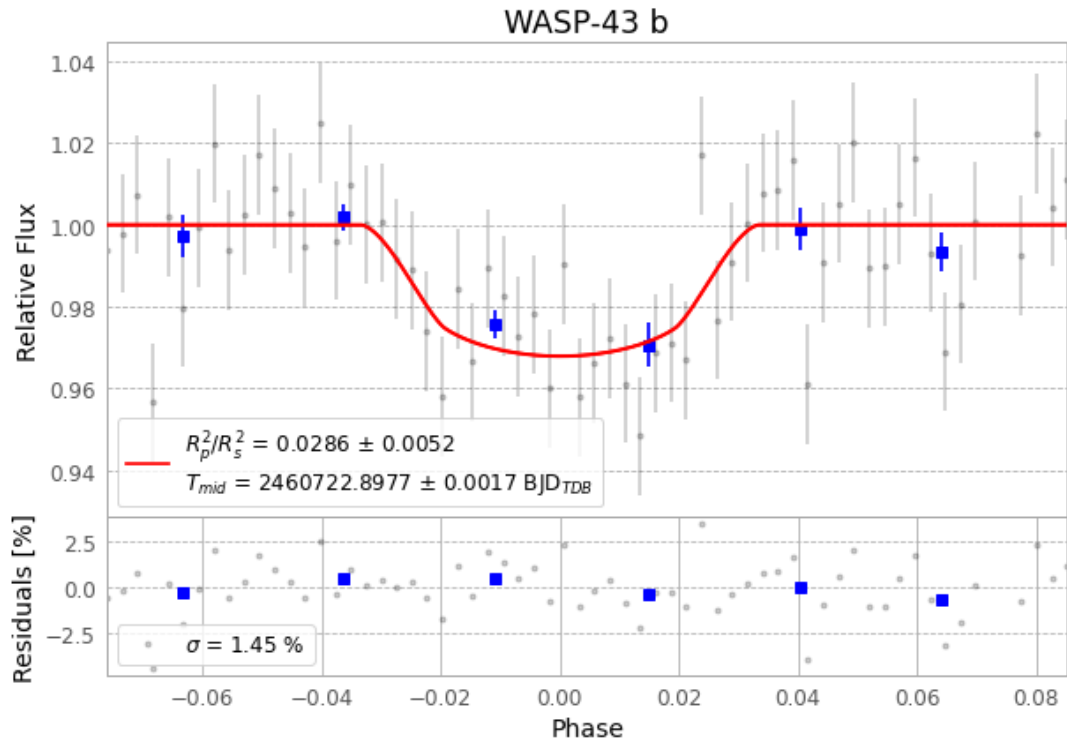


Figure 1 — FinalLightCurve_WASP-43 b_2025-02-16.png (SHA-256 68c8aaa3aa5a2273...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [394, 264], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 80a1d77ee7efbbfd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

64 FITS frames (42 MB), WASP-43250216075416.FITS → WASP-43250216110317.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-250216.log (88896e49100d...), FinalLightCurve_WASP-43 b_2025-02-16.png (68c8aaa3aa5a...), FinalLightCurve_WASP-43 b_2025-02-16.csv (3e4dd73bb656...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 04:19Z.